



Fatima Jinnah Women University

Department of Software Engineering

ASSIGNMENT No:01

Cloud Computing



NAME	Areesha Tasawar
REG. NO.	Fa24b1-se-047
SUBMITTED TO	Sir Waqas Saleem
SEMESTER	BSE-V(A)

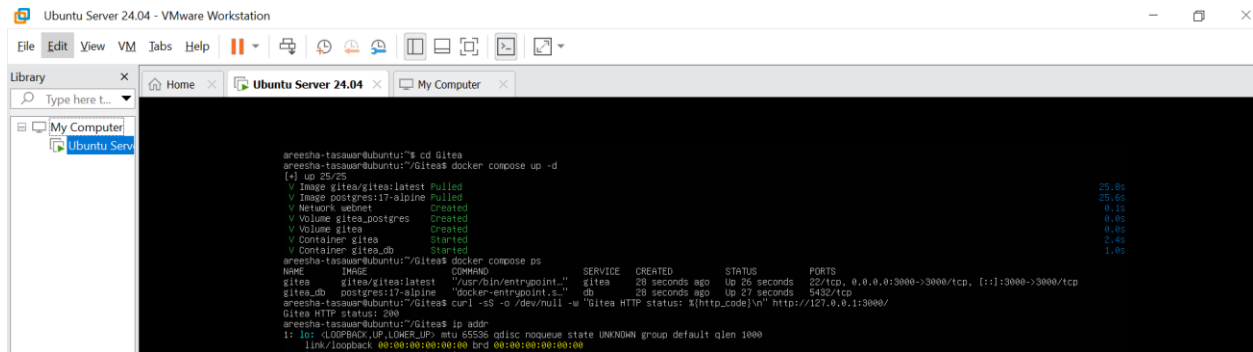
Assignment 01:

Git, Gitea, GitHub, Git LFS, and GitHub Pages

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

In this task, Docker Engine and Docker Compose were installed on the Ubuntu server. A containerized Gitea and PostgreSQL instance was successfully deployed, and a local repository was initialized and pushed to the running Gitea server.

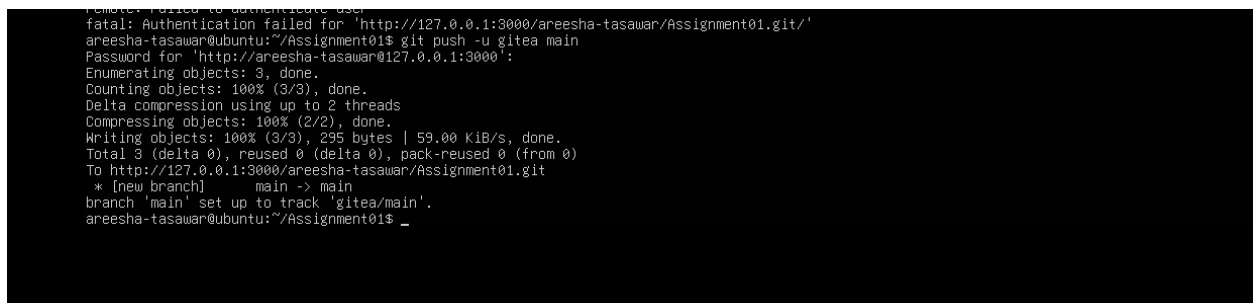
Step 1.1: Install and Run Gitea on the Ubuntu Server



```
areesha-tasawar@ubuntu:~/gitea$ cd gitea
areesha-tasawar@ubuntu:~/gitea$ docker compose up -d
[*] Up 25/25
✓ Image gitea/gitea:latest Pulled 25.8s
✓ Image postgres:17-alpine Pulled 25.6s
✓ Network webnet Created 0.1s
✓ Volume gitea_postgres Created 0.0s
✓ Volume gitea Created 0.0s
✓ Container gitea Started 2.4s
✓ Container gitea_db Started 1.0s
areesha-tasawar@ubuntu:~/gitea$ docker compose ps
NAME                IMAGE                       COMMAND                  SERVICE    CREATED      STATUS      PORTS
gitea                gitea/gitea:latest         "/usr/bin/entrypoint.sh" gitea      28 seconds ago Up 26 seconds 22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db             postgres:17-alpine         "docker-entrypoint.sh"  db         28 seconds ago Up 27 seconds 5432/tcp
areesha-tasawar@ubuntu:~/gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
areesha-tasawar@ubuntu:~/gitea$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
```

Figure1:task1_gitea_running.png

Step 1.2: Create and Push the Initial Repository



```
areesha-tasawar@ubuntu:~/Assignment01$ git init
fatal: Authentication failed for 'http://127.0.0.1:3000/areesha-tasawar/Assignment01.git/'
areesha-tasawar@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://areesha-tasawar@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 295 bytes | 59.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/areesha-tasawar/Assignment01.git
 * [new branch]    main -> main
branch 'main' set up to track 'gitea/main'.
areesha-tasawar@ubuntu:~/Assignment01$ _
```

Figure2:task1_gitea_push.png

Step 1.3: Verify the Gitea Repository

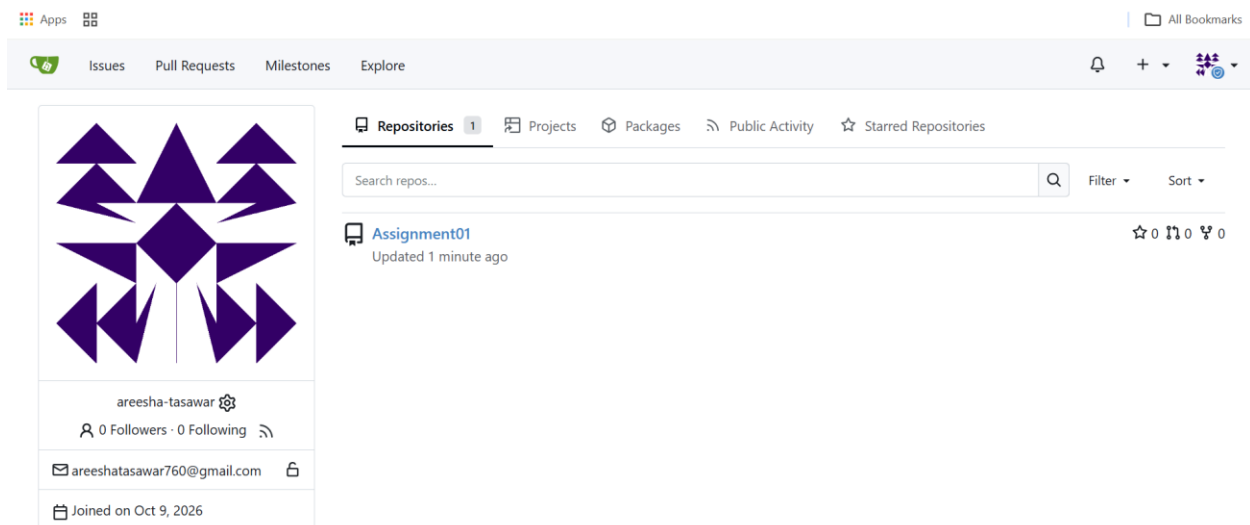


Figure3:task1_gitea_repository.png

Task 2: Push the Same Repository to GitHub

This task involved configuring dual remote URLs for the local Git repository on the Ubuntu server. The setup enables simultaneous code pushes and synchronization to both the self-hosted Gitea server and GitHub using secure SSH/HTTPS remotes.

Step 2.1: Create the GitHub Repository and Verify Both Remotes

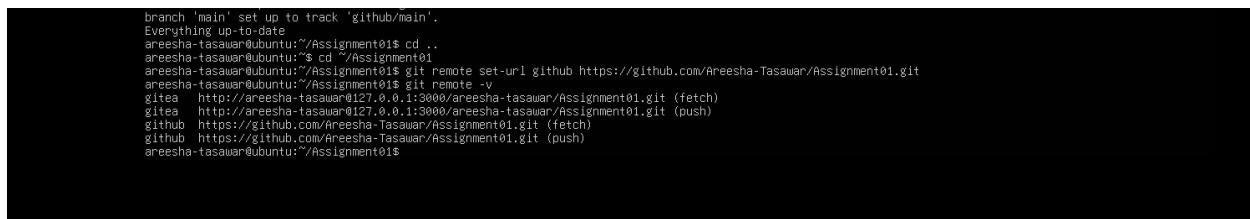


Figure4:task2_remotes.png

Step 2.2: Push to GitHub

```
Everything up-to-date
areesha-tasawar@ubuntu:~/Assignment01$ cd ..
areesha-tasawar@ubuntu:~/Assignment01$ git remote set-url github https://github.com/Areesha-Tasawar/Assignment01.git
areesha-tasawar@ubuntu:~/Assignment01$ git remote -v
github    https://github.com/Areesha-Tasawar/Assignment01.git (fetch)
github    https://github.com/Areesha-Tasawar/Assignment01.git (push)
areesha-tasawar@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': Areesha-Tasawar
Password for 'https://github.com':
branch 'main' set up to track 'github/main'.
Everything up-to-date
areesha-tasawar@ubuntu:~/Assignment01$
```

Figure5:task2_github_push.png

Step 2.3: Verify the GitHub Repository

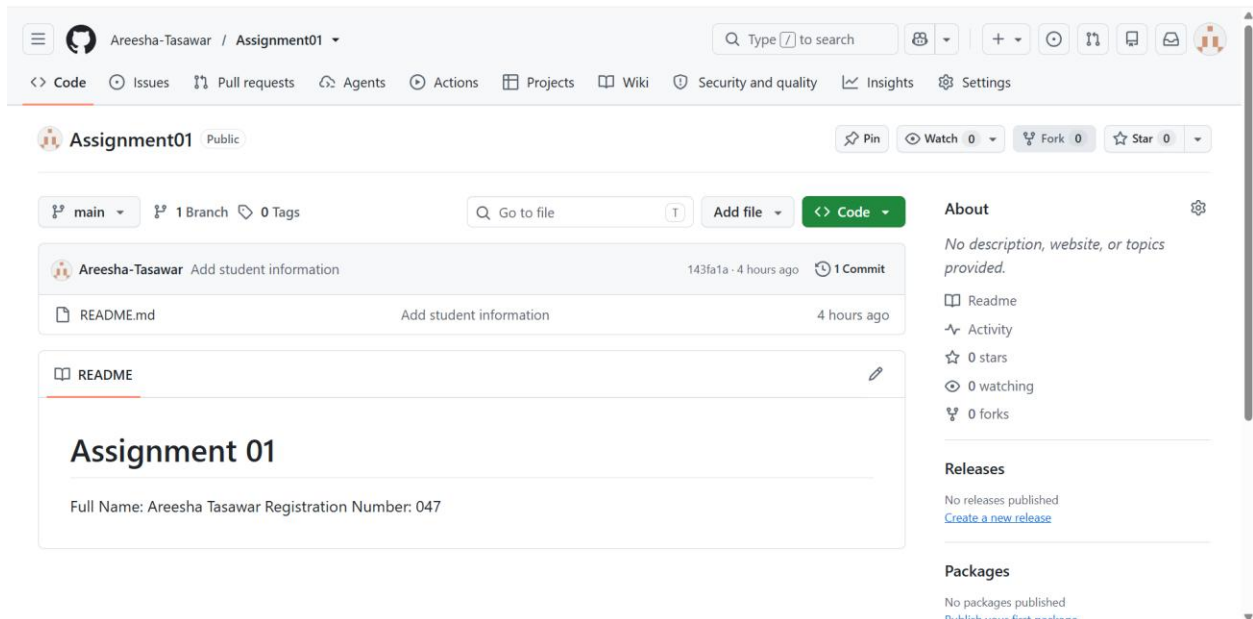
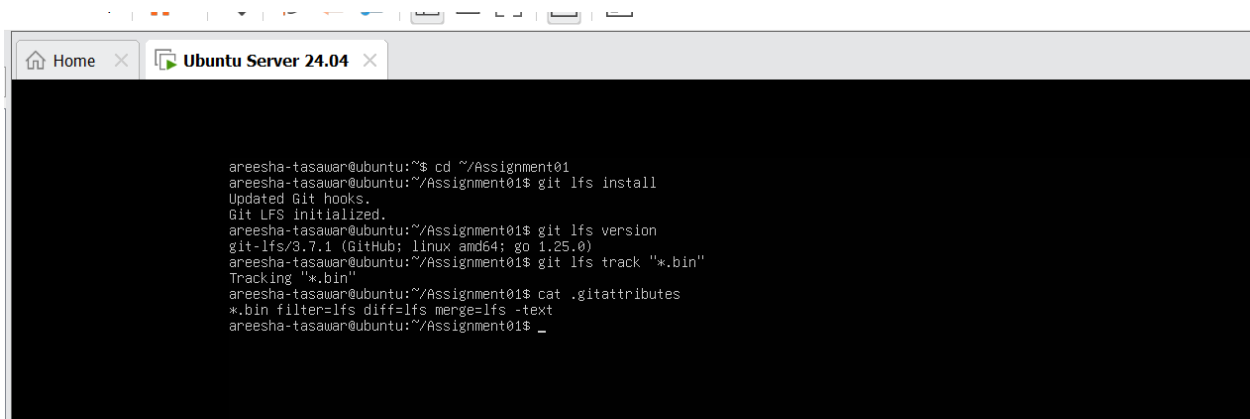


Figure6:task2_github_repository.png

Task 3: Track Three Large Files with Git LFS

Git LFS was initialized on the server repository to efficiently handle large binary assets (.bin files). Large files were successfully tracked, committed, and pushed to the remote repository without bloating the core Git history.

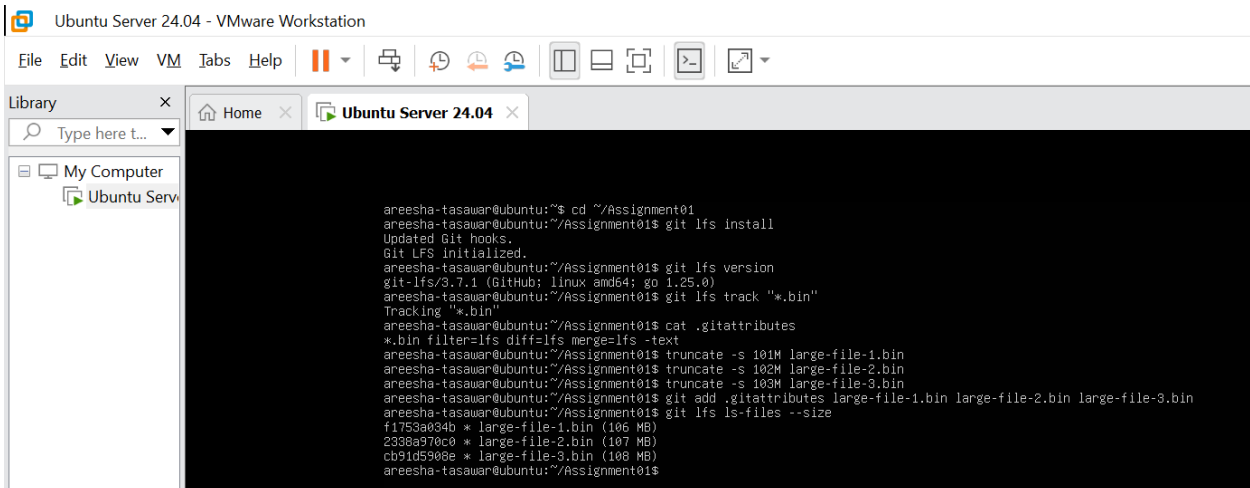
Step 3.1: Configure Git LFS



```
areesha-tasawar@ubuntu:~$ cd ~/Assignment01
areesha-tasawar@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
areesha-tasawar@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
areesha-tasawar@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
areesha-tasawar@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
areesha-tasawar@ubuntu:~/Assignment01$ _
```

Figure7:task3_lfs_setup.png

Step 3.2: Create and Verify Three LFS Files



```
areesha-tasawar@ubuntu:~$ cd ~/Assignment01
areesha-tasawar@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
areesha-tasawar@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
areesha-tasawar@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
areesha-tasawar@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
areesha-tasawar@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
areesha-tasawar@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
areesha-tasawar@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
areesha-tasawar@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
areesha-tasawar@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2330a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
areesha-tasawar@ubuntu:~/Assignment01$
```

Figure8:task3_lfs_files.png

Step 3.3: Commit and Push the LFS Files

```
areesha-tasawar@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (100 MB)
areesha-tasawar@ubuntu:~/Assignment01$ git commit -m "Add three files using Git LFS"
[main 4148eea] Add three files using Git LFS
4 files changed, 10 insertions(+)
create mode 100644 .gitattributes
create mode 100644 large-file-1.bin
create mode 100644 large-file-2.bin
create mode 100644 large-file-3.bin
areesha-tasawar@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': Areesha-Tasawar
Password for 'https://github.com': 
Username for 'https://github.com': Areesha-Tasawar
Password for 'https://github.com': 
Uploading LFS objects: 100% (3/3), 321 MB | 3.8 MB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 800 bytes | 400.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Areesha-Tasawar/Assignment01.git
 143fa1a..4148eea main -> main
areesha-tasawar@ubuntu:~/Assignment01$
```

Figure9:task3_lfs_push.png

Task 4: Create a Portfolio or CV with GitHub Pages

A personal portfolio website featuring About Me, Education, Skills, and Projects sections was developed using HTML and CSS. The source files were pushed to a public GitHub repository and published live using GitHub Pages.

Step 4.1: Create and Push the Portfolio Repository

The screenshot shows the GitHub repository page for 'Areesha-Tasawar' (Areesha-Tasawar.github.io). The repository is public and contains two files: 'index.html' and 'styles.css', both added 10 minutes ago. The repository has 1 commit. The right sidebar shows the 'About' section with no description, 0 stars, 0 watching, and 0 forks. The 'Releases' section shows no releases published. The 'Deployments' section shows 1 deployment using 'github-pages'.

Figure10:task4_pages_repository.png

Step 4.2: Publish with GitHub Pages

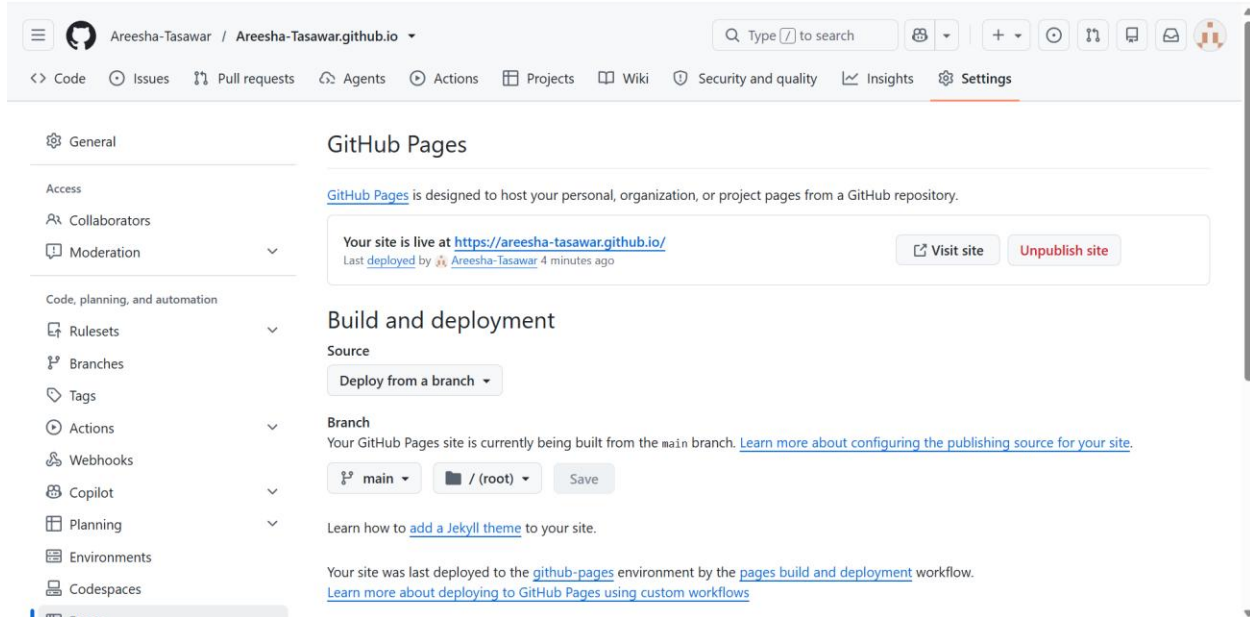


Figure11:task4_pages_deployment.png

Step 4.3: Verify the Live Portfolio

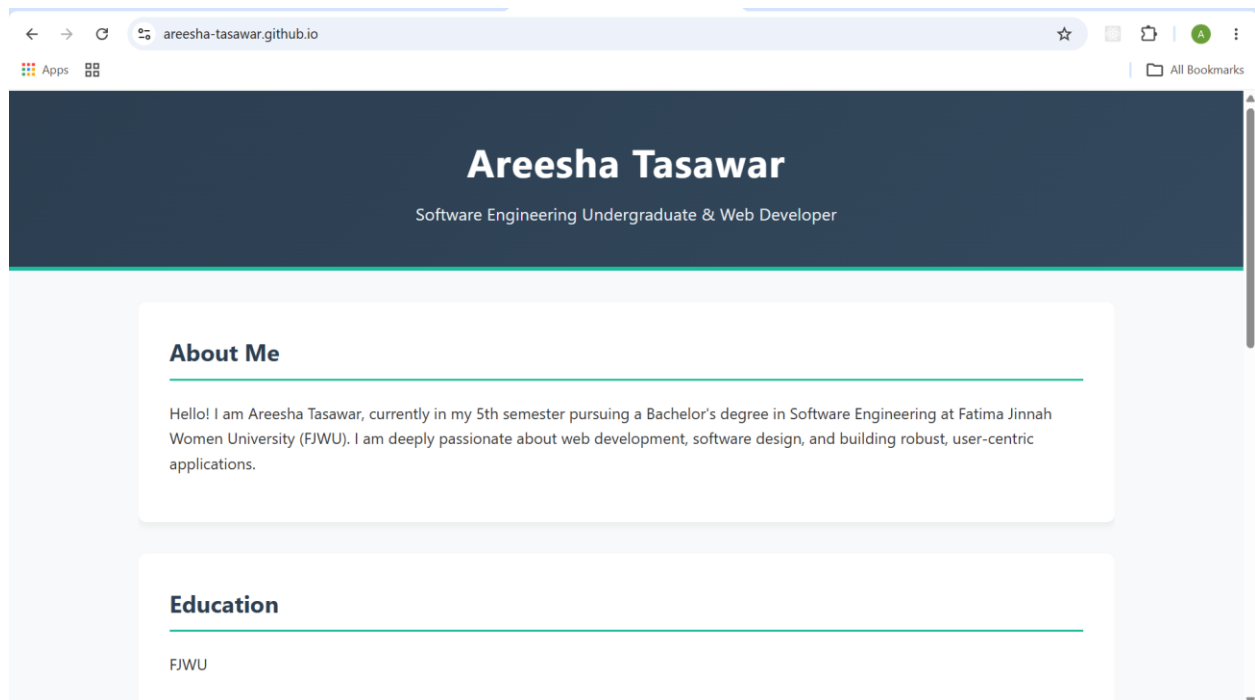


Figure12:task4_portfolio_live.png

Conclusion

This assignment successfully demonstrated essential software engineering and version control practices using an Ubuntu server environment. Key competencies achieved include self-hosting a version control service with Gitea, managing multi-remote repositories for synchronized code deployment, implementing Git Large File Storage (LFS) for efficient handling of large binary assets, and deploying a responsive personal portfolio using GitHub Pages. These hands-on tasks reinforced a comprehensive understanding of modern DevOps workflows, collaborative development, and professional deployment standards.